Free fall

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Describe and calculate ideal free fall near Earth.

SUCCESS CRITERIA - I can...

1. State the no-air-resistance assumption.
2. Use equal velocity changes in equal times.
3. Separate acceleration from mass and current velocity.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A simplified model predicts position (-40, 60) km with bounds ±4 km in x and ±6 km in y. Write a short briefing giving both intervals and two reasons it is not a real asteroid collision forecast.

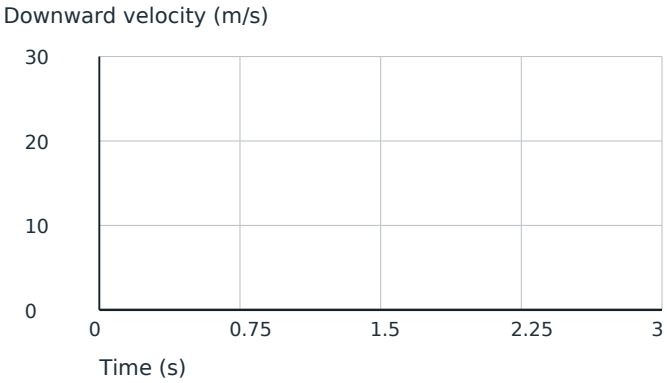
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $v_f = v_i + g\,\Delta t$; $\Delta y = v_i\,\Delta t + \frac{1}{2}g(\Delta t)^2$. Downward positive for these equations; g = 10 m/s². Ideal free fall: gravity is the only force.

Worked reasoning: A ball dropped from rest has downward speeds 0, 10, 20 m/s at 0, 1, 2 s in ideal free fall with g=10 m/s². Equal times give equal velocity changes.

Your turn: Sketch its velocity-time graph with downward positive. Explain why the positions are not equally spaced.



Marzano check - circle: ① ② ③

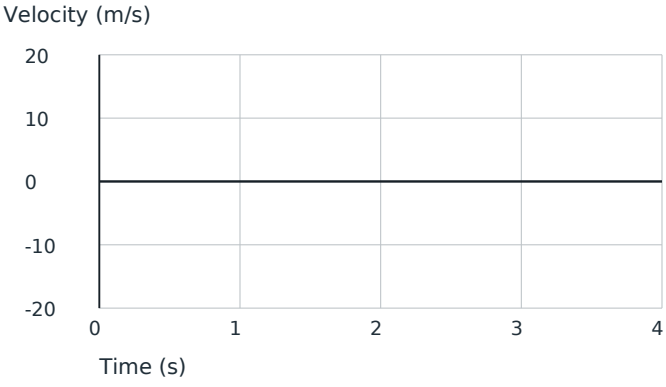
03 Read and interpret the model

Two balls of different mass are released together in a vacuum from the same height. Compare acceleration and arrival time.

Marzano check - circle: ① ② ③

04 Apply the physics

A ball is thrown straight upward at 20 m/s. Take upward as positive; use g = 10 m/s² and ignore air resistance. Complete its velocity-time graph. When does it reach the top, and what is its velocity at 3 s?



Marzano check - circle: ① ② ③

05 Explain your reasoning

At the highest point of a tossed ball, a classmate says gravity stops because speed is zero. Correct the claim.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

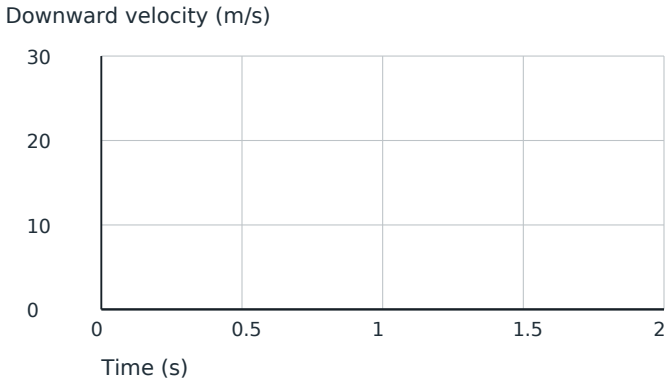
An a-versus-F_{net} graph has slope 2 (m/s²)/N. Find mass and acceleration at 3 N. Explain why a driving-force plot can have an offset.
Use the table to keep the controlled and changed quantities distinct.

Trial	Mass	Net force	Acceleration
A	____	____	____
B	____	____	____

Marzano check - circle: 1 2 3

07 Exit check

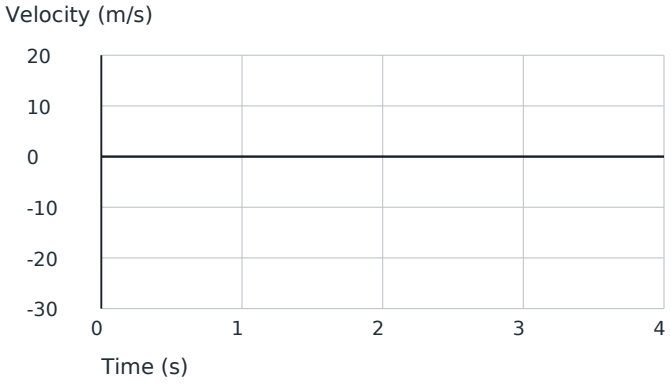
A ball starts downward at 5 m/s. It falls for 2 s with g = 10 m/s² and negligible air resistance. Draw its velocity-time graph. Use slope and area to find its final velocity and downward displacement.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A ball is thrown up at 15 m/s from a 20 m platform. Use g=10 m/s². Find its height at 2 s and its time to reach the ground.



Marzano check - circle: 1 2 3